

Ocean Data API Documentation

ENDPOINT URL (Water Quality Now Cast)

<https://gemini.incois.gov.in/OceanDataAPI/api/wqns/{station}/{parameter}>

Description

The Ocean Data API allows users to retrieve oceanographic data for specified parameters from monitoring stations at Kochi and Vizag. The data includes various water quality parameters essential for scientific research, environmental monitoring, and maritime operations.

Endpoints

GET /wqns/{station}/{parameter}

This endpoint retrieves the latest value of the specified parameter from the given station. An API key is required for authorization.

Path Parameters

- **station** (string): The name of the monitoring station from which data is being requested.
Valid values:
 - Kochi
 - Vizag
- **parameter** (string): The oceanographic or water quality parameter you wish to retrieve.
Valid values:
 - **currentspeed**: Current speed
 - **currentdirection**: Current Direction
 - **ph**: pH level of the water.
 - **salinity**: Salinity level of the water.
 - **temperature**: Temperature of the water.
 - **dissolvedoxygen**: Concentration of dissolved oxygen in the water.
 - **dissolvedmethane**: Concentration of dissolved methane in the water.
 - **pco2air**: Partial pressure of CO2 in the air.
 - **pco2water**: Partial pressure of CO2 in the water.
 - **chlorophyll**: Chlorophyll concentration in the water.
 - **phycocyanin**: Phycocyanin concentration, indicating the presence of cyanobacteria.

- **phycoerythrin:** Phycoerythrin concentration, indicating the presence of certain algae.
- **turbidity:** Turbidity of the water, a measure of how clear or cloudy the water is.
- **scattering:** Scattering coefficient, indicating particles suspended in the water.
- **cdom:** Concentration of coloured dissolved organic matter (CDOM) in the water.

Responses

- **200 OK**

- The request was successful, and the data for the specified parameter from the given station is returned.

Example Response:

```
{
  "metadata": {
    "station_name": "Vizag",
    "LastReported": "2024-06-07 05:30:56",
    "latitude": 17.712965,
    "longitude": 83.3807068,
    "parameter": "pco2air",
    "units": "µatm"
  },
  "observationTime": [
    "2024-06-07 05:30:56",
    "2024-06-07 05:00:57",
    "2024-06-07 04:30:56"
  ],
  "pco2air": [
    "400.0000000",
    "400.0000000",
    "400.0000000"
  ]
}
```

- **400 Bad Request**

- The server could not understand the request due to incorrect syntax. This might occur if an invalid station or parameter is provided.

- **404 Not Found**

- The specified station or parameter could not be found.

- **500 Internal Server Error**

- The server encountered an unexpected condition that prevented it from fulfilling the request.

Example Code in JS:

```
async function fetchAPIData() {
  const url = "APIURL";
  const response = await fetch(url, {
    'headers': { 'Authorization': 'API_ACCESS_KEY' },
  });
  if (response.status === 200) {
    const responseJson = await response.json();
    console.log(responseJson);
    console.log(typeof(responseJson));
  } else {
```